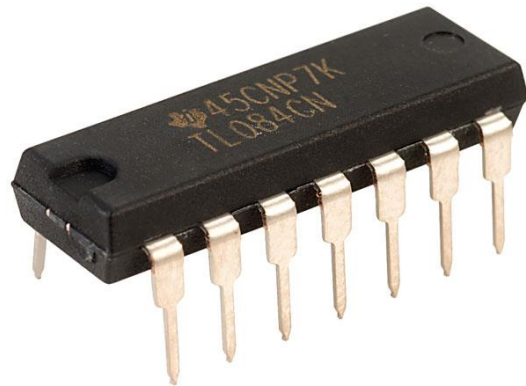
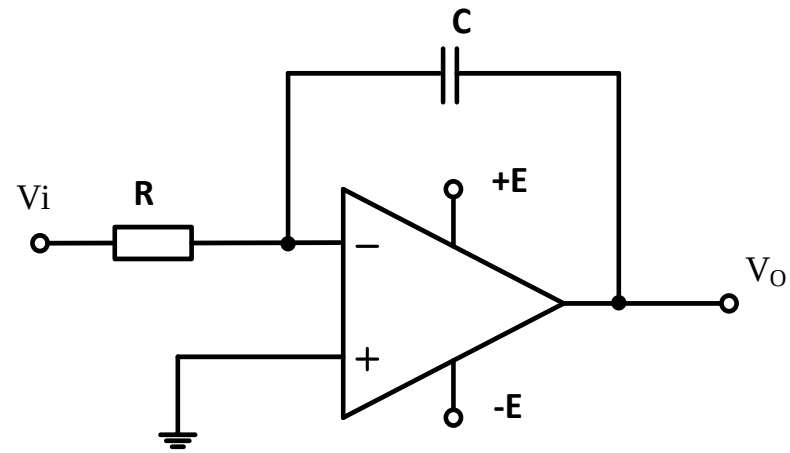


Bazele Circuitelor Electronice

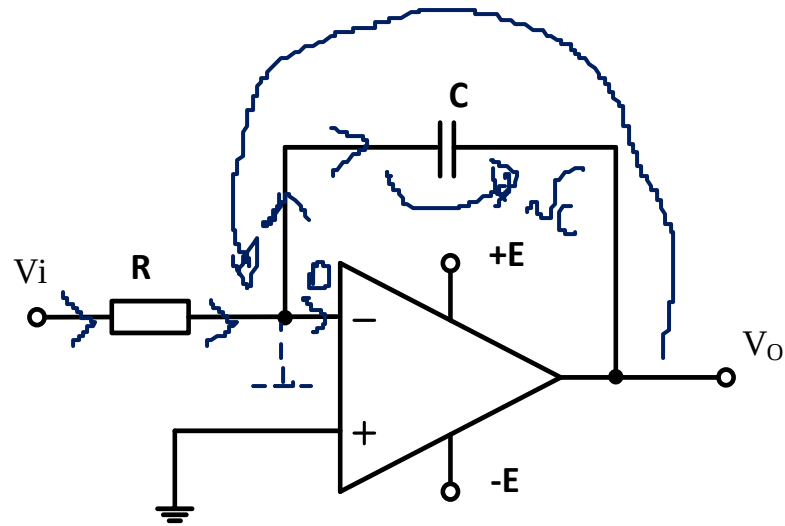
Aplicații cu Amplificatoare Operaționale



Integrator



Integrator



$$\underbrace{R \parallel \infty \Rightarrow v^+ = v^-}_{v^+ = 0 = v^-} \left\{ \Rightarrow v^- = 0 \right.$$

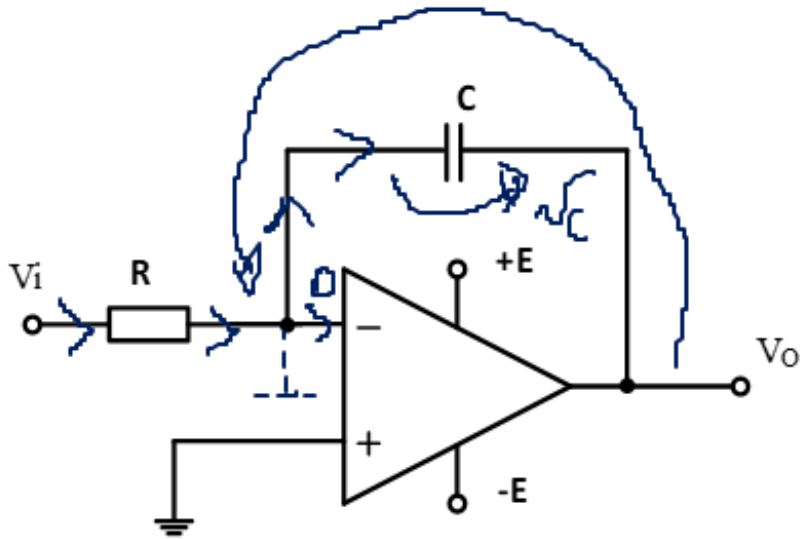
$$\underbrace{i_R = i_C}_{i_R = i_C} \rightarrow i_R = \frac{v_i - 0}{R} = \frac{v_i}{R}$$

$$\underbrace{v_o = -v_c}_{v_o = -v_c} \quad i_C = C \frac{dv_c}{dt} \quad | \quad \int$$

$$\Rightarrow v_o = -\frac{1}{C} \int i_c dt$$



Integrator



$$v_c = \frac{1}{C} \int i_c dt$$

$$i_c = i_R = \frac{v_i}{R}$$

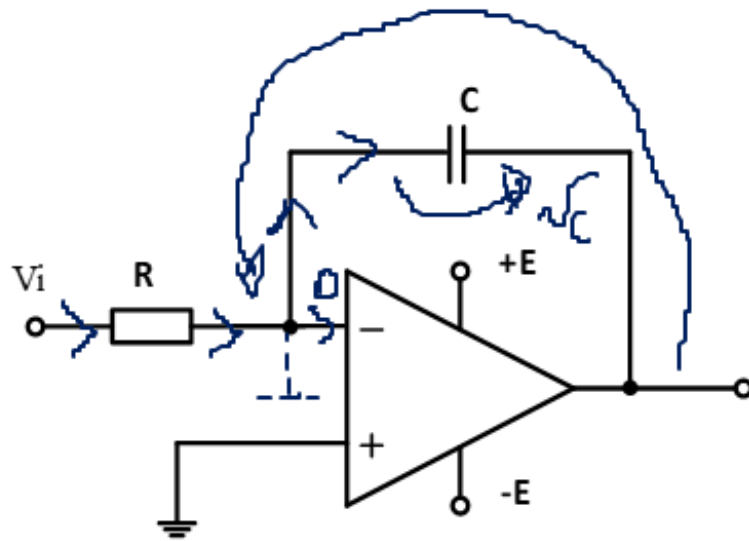
$$\Rightarrow v_c = \frac{1}{C} \int \frac{v_i}{R} dt$$

$$v_c = \frac{1}{RC} \int v_i(t) dt$$

$$v_o = -v_c$$

$$v_o = -\frac{1}{RC} \int v_i(t) dt$$

Integrator



$$RN \Rightarrow v^+ = v^-$$

$$v^+ = 0$$

$$\Rightarrow v^- = 0 \text{ (masă virtuală)}$$



$$i_R = \frac{v_i - 0}{R} = \frac{v_i}{R}$$

$$i_C = C \frac{dv_C}{dt} \Rightarrow v_C = \frac{1}{C} \int i_C dt$$

Curenți de polarizare nuli

$$\Rightarrow i_R = i_C$$

$$i_R = \frac{v_i - 0}{R} = \frac{v_i}{R}$$

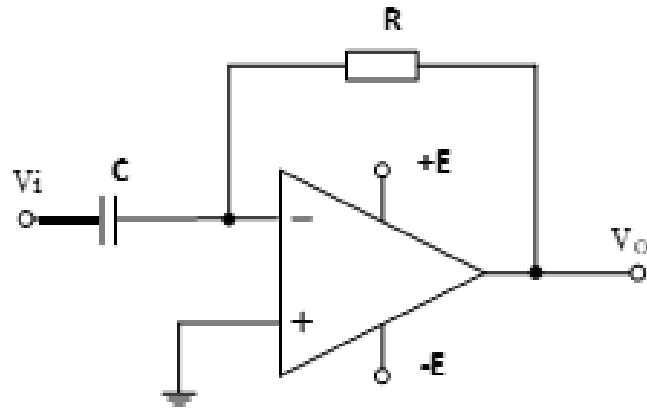
$$v_o = -v_C$$

$$v_o = -v_C = -\frac{1}{C} \int i_C dt = -\frac{1}{C} \int i_R dt$$

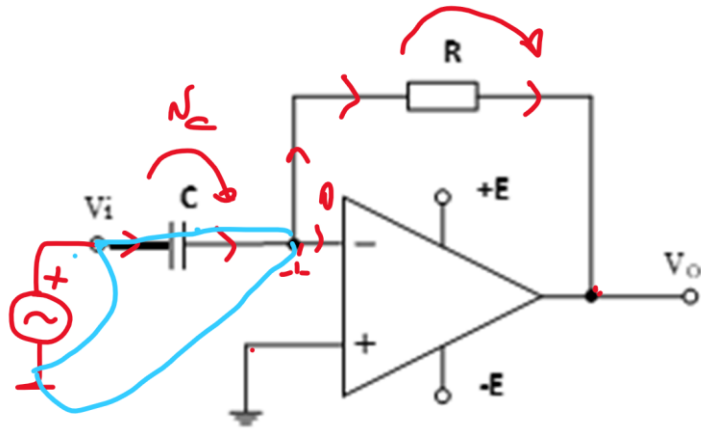
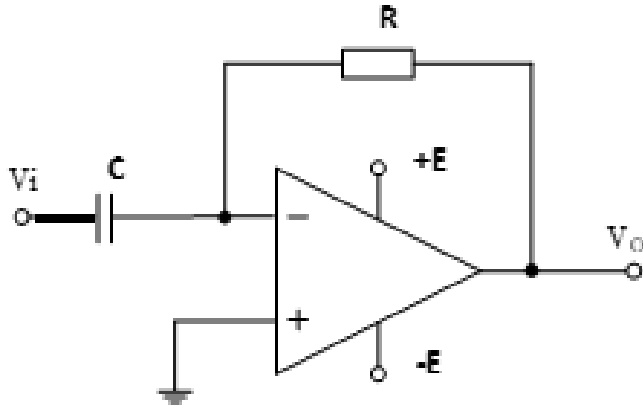
$$v_o = -\frac{1}{C} \int \frac{v_i}{R} dt = -\frac{1}{RC} \int v_i dt$$

$$v_o = -\frac{1}{RC} \int v_i dt$$

Derivator



Derivator



$$i_c = i_R$$

$$V_R = -V_R$$

$$V_R = R \times i_R$$

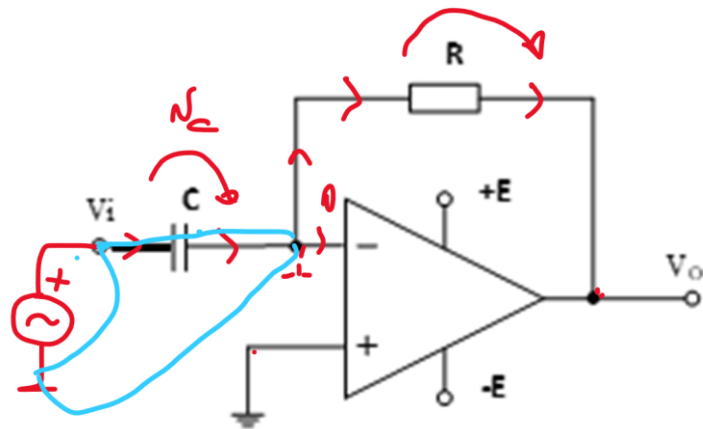
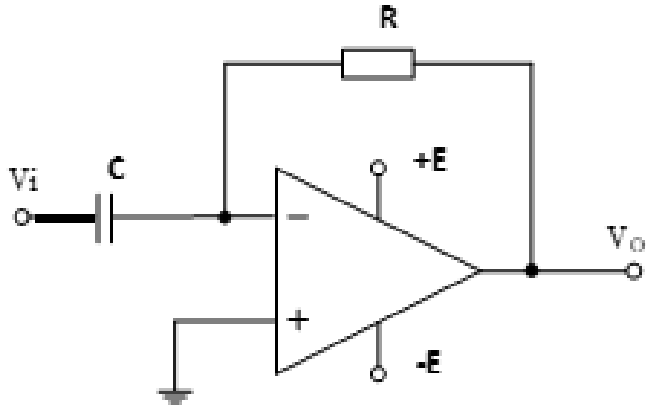
$$V_o = -V_R = -R \cdot i_R$$



$$V_o = -R \times i_c = -R C \frac{dV_i}{dt} = -RC \frac{dV_i}{dt}$$

$$V_o = -RC \frac{dV_i}{dt}$$

Derivator



$$i_C = i_R$$

$$v_O = -v_R$$

$$v_R = R \cdot i_R$$

$$\Rightarrow v_O = -v_R = -R \cdot i_R$$

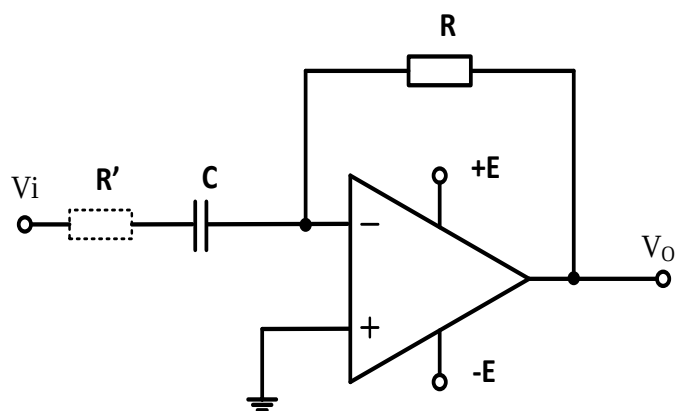
$$i_C = C \frac{dv_C}{dt}$$

$$v_O = -R \cdot i_R = -R \cdot i_C = -RC \frac{dv_C}{dt} = -RC \frac{dv_i}{dt}$$

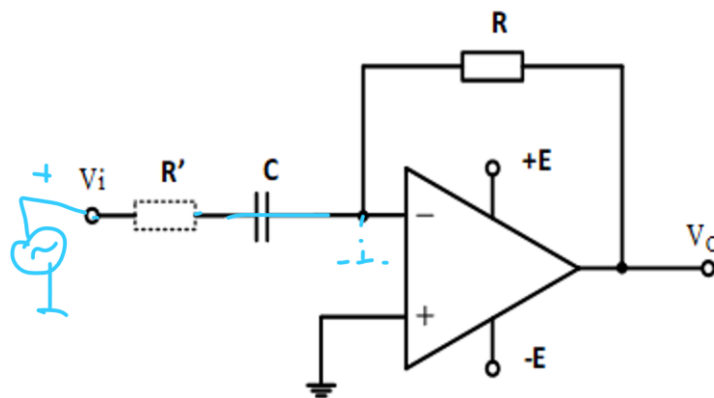
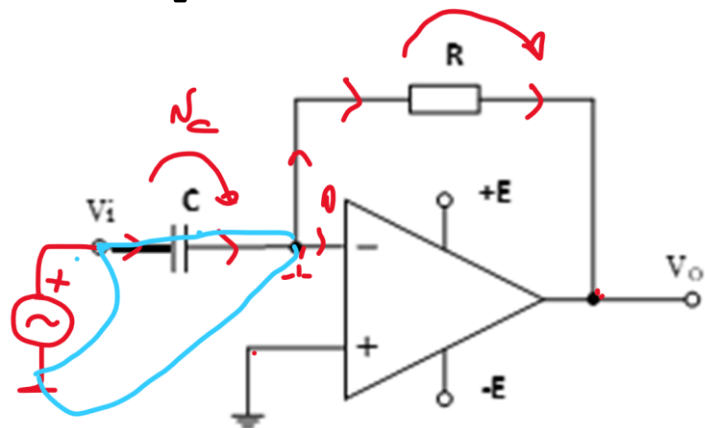
$$v_O = -RC \frac{dv_i}{dt}$$

Derivator

La frecvențe mari reactanța condensatorului tinde la 0 (scurtcircuit).
Pentru a evita acest scurtcircuit al sursei, la frecvențe mari se introduce rezistența R' .



$$Z_C = \frac{1}{j\omega C} = \frac{1}{j2\pi f C} \Rightarrow Z_C \rightarrow 0 \text{ as } f \rightarrow \infty$$

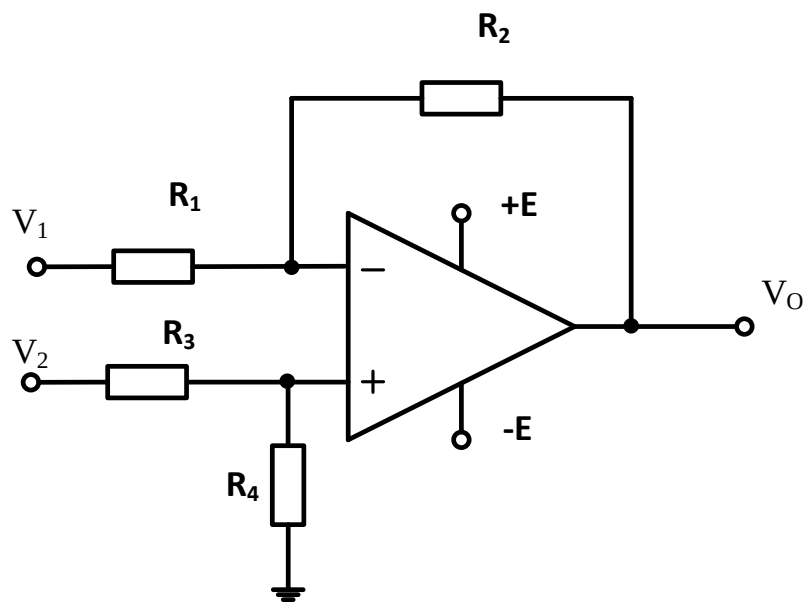


$$f \rightarrow \infty \Rightarrow Z_C \rightarrow 0$$

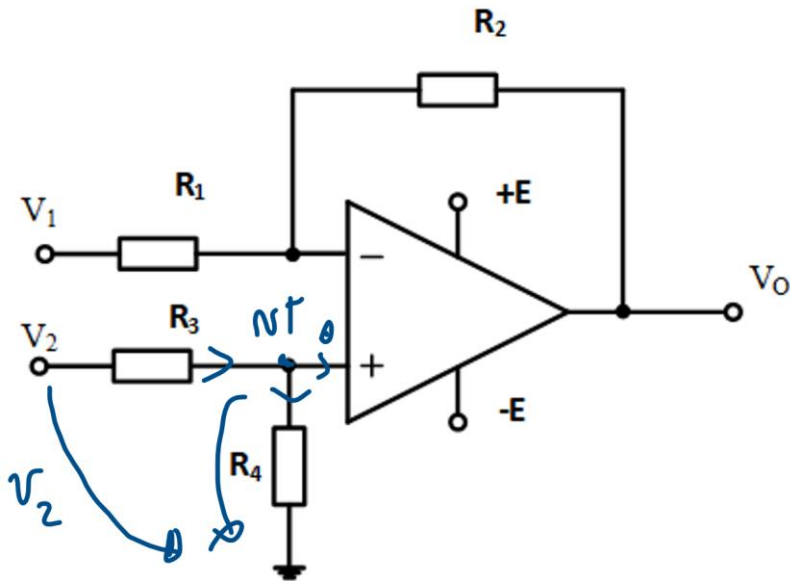
$$v_o = -RC \frac{dv_i}{dt} - \frac{R}{R'} \cdot v_i$$



Amplificator diferențial



Amplificator diferențial



$$v_o = f(v_1, v_2) = ?$$

$$RN \Rightarrow v^+ = v^-$$

$$v^+ = \frac{\frac{v_2}{R_3} + \frac{0}{R_4}}{\frac{1}{R_3} + \frac{1}{R_4}} = \frac{\frac{v_2}{R_3}}{\frac{R_3 + R_4}{R_3 R_4}} = \frac{R_4}{R_3 + R_4} v_2$$

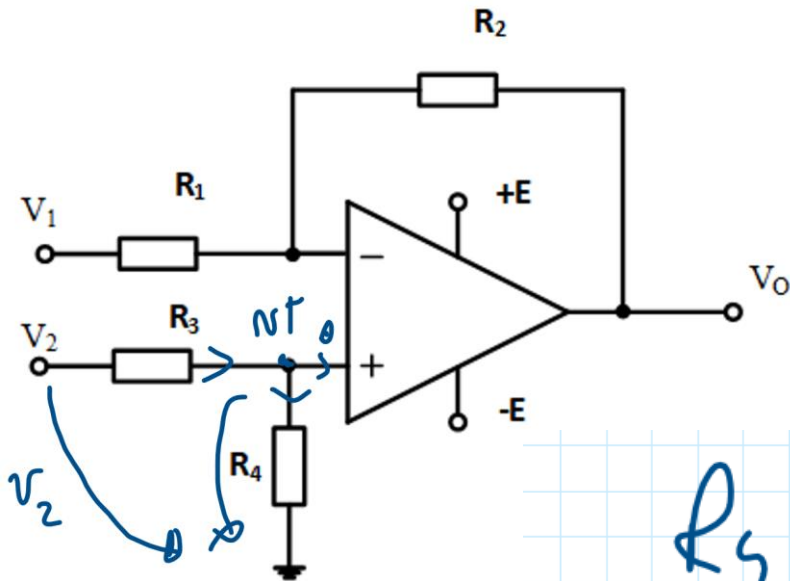
$$v^- = \frac{\frac{v_1}{R_1} + \frac{v_o}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{\frac{R_1 R_2}{R_1 + R_2} \left(\frac{R_2 v_1 + R_1 v_o}{R_1 R_2} \right)}{\frac{R_1 + R_2}{R_1 R_2}} = \frac{1}{R_1 + R_2} (R_2 v_1 + R_1 v_o)$$



Amplificator diferențial

$$v_o = f(v_1, v_2) = ?$$

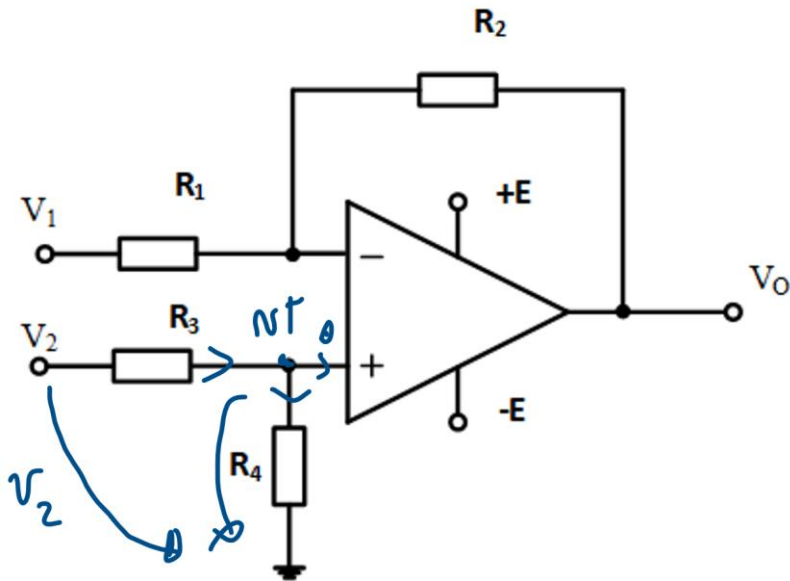
$$RN \Rightarrow v^+ = v^-$$



$$\frac{R_4}{R_3 + R_4} v_2 = \frac{R_2}{R_1 + R_2} v_1 + \frac{R_1}{R_1 + R_2} v_o$$



Amplificator diferențial



$$v_o = f(v_1, v_2) = ?$$

$$RN \Rightarrow v^+ = v^-$$

$$\frac{R_1}{R_3 + R_1} v_2 = \frac{R_2}{R_1 + R_2} v_1 + \frac{R_1}{R_1 + R_2} v_o$$

$$\text{pt. } \underline{R_1 = R_3} \quad \text{și} \quad \underline{R_2 = R_4}$$

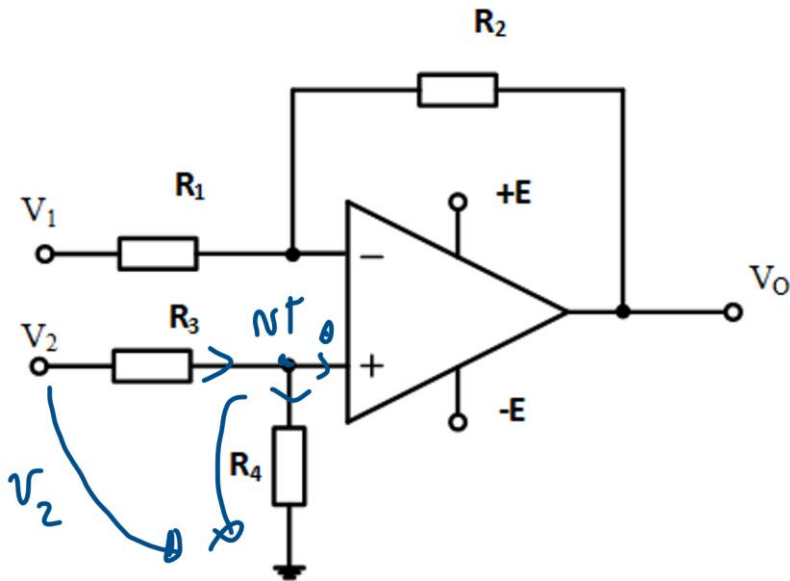
$$\Rightarrow v_o = \frac{R_1 + R_2}{R_1 + R_2} \left(\frac{R_2}{R_1} v_2 - \frac{R_2}{R_1} v_1 \right)$$

$$v_o = \frac{R_2}{R_1} (v_2 - v_1)$$



Amplificator diferențial

$$v_o = f(v_1, v_2)$$



$$RN \Rightarrow v^+ = v^-$$

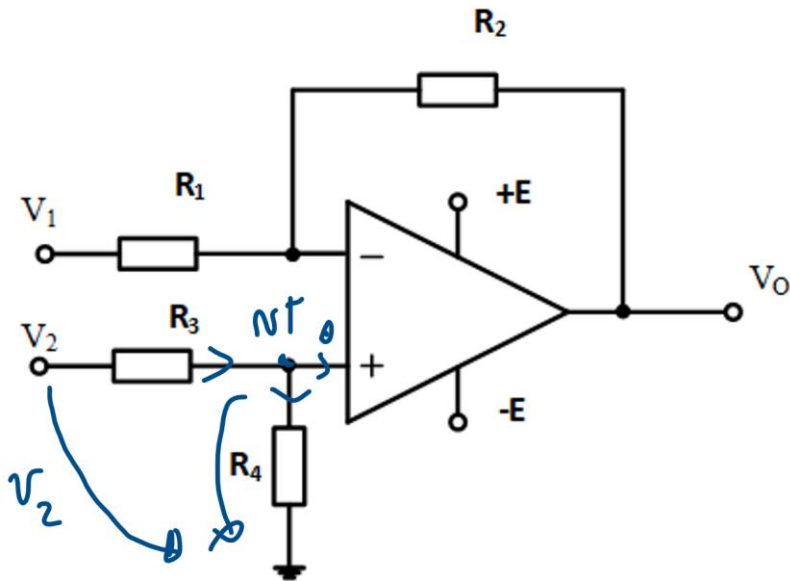
$$v^+ = \frac{\frac{v_2}{R_3} + \frac{0}{R_4}}{\frac{1}{R_3} + \frac{1}{R_4}} = \frac{\frac{v_2}{R_3}}{\frac{R_3 + R_4}{R_3 \cdot R_4}} = \frac{R_4}{R_3 + R_4} \cdot v_2$$

$$v^- = \frac{\frac{v_1}{R_1} + \frac{v_o}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{R_1 \cdot R_2}{R_1 + R_2} \cdot \frac{R_2 v_1 + R_1 v_o}{R_1 \cdot R_2} = \frac{1}{R_1 + R_2} \cdot (R_2 v_1 + R_1 v_o)$$



Amplificator diferențial

$$v_o = f(v_1, v_2)$$



$$RN \Rightarrow v^+ = v^-$$

$$\Rightarrow \frac{R_4}{R_3 + R_4} \cdot v_2 = \frac{R_2}{R_1 + R_2} \cdot v_1 + \frac{R_1}{R_1 + R_2} \cdot v_o$$

pentru $R_1 = R_3$ și $R_2 = R_4$

$$\Rightarrow v_o = \frac{R_1 + R_2}{R_1 + R_2} \left(\frac{R_2}{R_1} v_2 - \frac{R_2}{R_1} v_1 \right)$$

$$v_o = \frac{R_2}{R_1} (v_2 - v_1) = -\frac{R_2}{R_1} (v_1 - v_2)$$

